

Lab 1-2 VLAN Routing

Learning Objectives

As a result of this lab section, you should achieve the following tasks:

- Establishment of a trunk interface for VLAN routing.
- Configuration of sub-interfaces on a single physical interface.
- Enabling of ARP messages to be broadcast between VLANs.

Topology

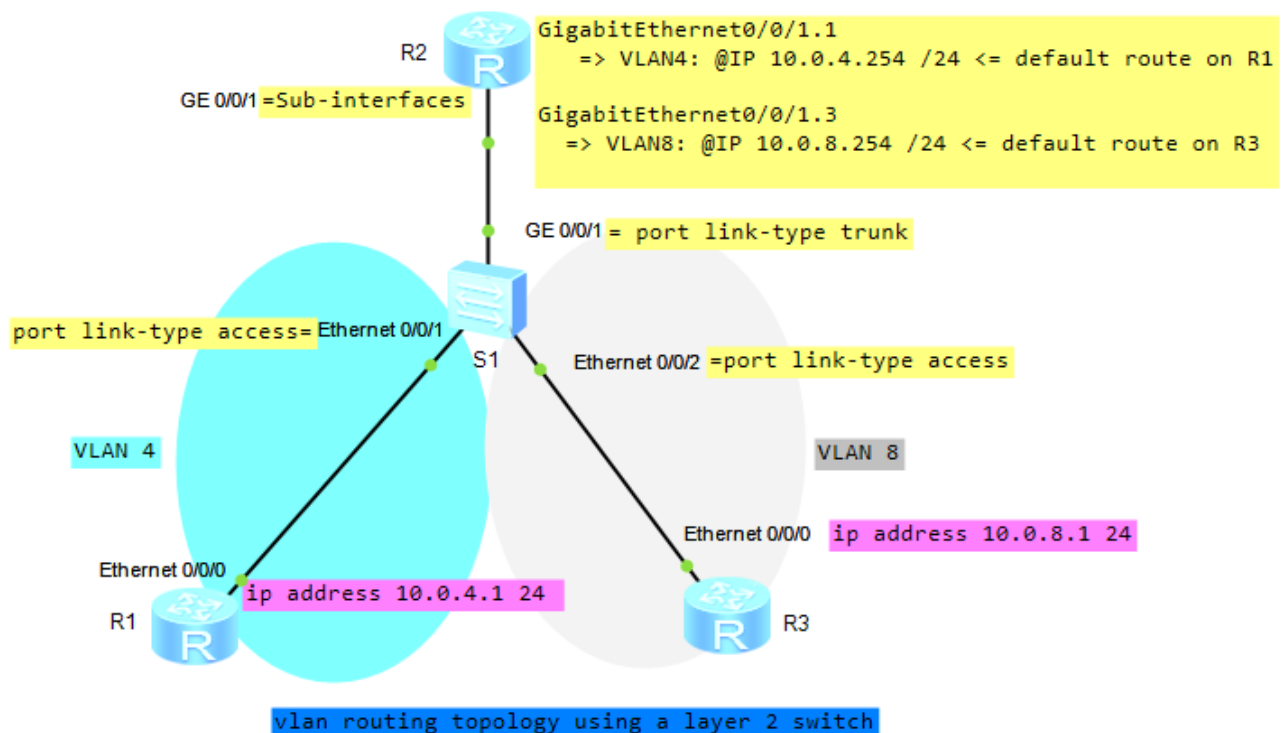


Figure 1. VLAN routing topology using a layer 2 switch.

Scenario

The implementation of VLANs in the enterprise network has resulted in groups of users being isolated from other users that are part of different subnets. As the network administrator you have been given the task to ensure that the broadcast domains are maintained whilst allowing communication between the disparate users.

Tasks

Step 1 Preparing the environment

If you are starting this section with a non-configured device, begin here and then move to step 3. For those continuing from previous labs, begin at step 2.

Configure the system name for R1, R3 and S1. Configure the IP address 10.0.4.1/24 on interface Ethernet 0/0/0.

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]interface Ethernet 0/0/0
[R1-Ethernet0/0/0]ip address 10.0.4.1 24
```

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
```

```
<Quidway>system-view
[Quidway]sysname S1
```

Step 2 Clean up the previous configuration

Remove the IP address 10.0.4.3 from R3, and disable the switch interfaces between S1 and S3 and S2 and S4 respectively.

```
[R3]interface GigabitEthernet 0/0/2
[R3-GigabitEthernet0/0/2]undo ip address
```

```
[S1]undo gvrp
Warning: All information about the GVRP will be deleted . Continue?[Y/N]:y
Info: This operation may take a few seconds. Please wait for a moment...done.
[S1]interface GigabitEthernet 0/0/13
[S1-GigabitEthernet0/0/13]undo port trunk allow-pass vlan 2 to 4094
[S1-GigabitEthernet0/0/13]shutdown
[S1-GigabitEthernet0/0/13]quit
[S1]interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1]undo port hybrid vlan 2 4
[S1-GigabitEthernet0/0/1]quit
[S1]undo vlan batch 2 100 200
Warning: The configurations of the VLAN will be deleted. Continue?[Y/N]:y
Info: This operation may take a few seconds. Please wait for a moment...done.
```

```
[S2]undo gvrp
Warning: All information about the GVRP will be deleted . Continue?[Y/N]:y
Info: This operation may take a few seconds. Please wait for a moment...done.
[S2]interface GigabitEthernet 0/0/6
[S2-GigabitEthernet0/0/6]undo port trunk allow-pass vlan 2 to 4094
[S2-GigabitEthernet0/0/6]shutdown
[S2-GigabitEthernet0/0/6]quit
[S2]interface GigabitEthernet 0/0/3
[S2-GigabitEthernet0/0/3]undo port hybrid vlan 2 4
[S2-GigabitEthernet0/0/3]quit
[S2]undo vlan batch 2 100 200
Warning: The configurations of the VLAN will be deleted. Continue?[Y/N]:y
Info: This operation may take a few seconds. Please wait for a moment...done.
```

```
[S3]undo gvrp
Warning: All information about the GVRP will be deleted . Continue?[Y/N]:y
Info: This operation may take a few seconds. Please wait for a moment...done.
[S3]interface Ethernet 0/0/13
[S3-Ethernet0/0/13]undo port trunk allow-pass vlan 2 to 4094
[S3-Ethernet0/0/13]port link-type hybrid
[S3-Ethernet0/0/13]quit
[S3]interface Ethernet 0/0/1
[S3-Ethernet0/0/1]undo port trunk allow-pass vlan 2 to 4094
[S3-Ethernet0/0/1]quit
```

```
[S3]undo vlan 2
```

```
[S4]undo gvrp
```

Warning: All information about the GVRP will be deleted . Continue?[Y/N]:y

Info: This operation may take a few seconds. Please wait for a moment...done.

```
[S4]interface Ethernet 0/0/6
```

```
[S4-Ethernet0/0/6]undo port trunk allow-pass vlan 2 to 4094
```

```
[S4-Ethernet0/0/6]port link-type hybrid
```

```
[S4-Ethernet0/0/6]quit
```

```
[S4]interface Ethernet 0/0/1
```

```
[S4-Ethernet0/0/1]undo port trunk allow-pass vlan 2 to 4094
```

```
[S4-Ethernet0/0/1]quit
```

```
[S4]undo vlan 2
```

Step 3 **Configure an IP address for R3**

Configure an IP address in the 10.0.8.0/24 network range on R1 interface Gigabit Ethernet 0/0/1

```
[R3]interface Ethernet 0/0/0
```

```
[R3-Ethernet0/0/0]ip address 10.0.8.1 24
```

Step 4 **Establish two VLANs**

Create VLANs 4 and 8 on S1, configure interface Gigabit Ethernet 0/0/1 to belong to VLAN 4, and interface Gigabit Ethernet 0/0/3 to belong to VLAN 8.

```
[S1]vlan batch 4 8
```

Info: This operation may take a few seconds. Please wait for a moment...done.

```
[S1]interface Ethernet 0/0/1
```

```
[S1-Ethernet0/0/1]port link-type access
```

```
[S1-Ethernet0/0/1]port default vlan 4
```

```
[S1-Ethernet0/0/1]quit
```

```
[S1]interface Ethernet0/0/2
```

```
[S1-Ethernet0/0/2]port link-type access
```

```
[S1-Ethernet0/0/2]port default vlan 8
```

```
[S1-Ethernet0/0/2]quit
```

Set interface Gigabit Ethernet 0/0/2 as a trunk link for VLANs 4 and 8.

```
[S1]interface GigabitEthernet0/0/1
```

```
[S1-GigabitEthernet0/0/2]port link-type trunk
```

```
[S1-GigabitEthernet0/0/2]port trunk allow-pass vlan 4 8
```

Step 5 **Configure VLAN routing through the sub-interface of R2**

Configure sub-interfaces GigabitEthernet0/0/1.1 and GigabitEthernet0/0/1.3, to act as the gateway of VLAN 4, and act as the gateway of VLAN 8.

```
<Huawei>system-view
```

Enter system view, return user view with Ctrl+Z.

```
[Huawei]sysname R2
```

```
[R2]interface GigabitEthernet0/0/1.1
```

```
[R2-GigabitEthernet0/0/1.1]ip address 10.0.4.254 24
```

```
[R2-GigabitEthernet0/0/1.1]dot1q termination vid 4
```

```
[R2-GigabitEthernet0/0/1.1]arp broadcast enable
```

```
[R2-GigabitEthernet0/0/1.1]quit
```

```
[R2]interface GigabitEthernet0/0/1.3
```

```
[R2-GigabitEthernet0/0/1.3]ip address 10.0.8.254 24
```

```
[R2-GigabitEthernet0/0/1.3]dot1q termination vid 8
```

```
[R2-GigabitEthernet0/0/1.3]arp broadcast enable
```

Test connectivity between R1 and R3.

```
<R1>ping 10.0.8.1
PING 10.0.8.1: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out

--- 10.0.8.1 ping statistics ---
5 packet(s) transmitted
0 packet(s) received
100.00% packet loss
```

Configure a default route on R1 and R3.

```
[R1]ip route-static 0.0.0.0 0.0.0.0 10.0.4.254
[R3]ip route-static 0.0.0.0 0.0.0.0 10.0.8.254
```

Test connectivity between R1 and R3 again.

```
<R1>ping 10.0.8.1
PING 10.0.8.1: 56 data bytes, press CTRL_C to break
Reply from 10.0.8.1: bytes=56 Sequence=1 ttl=254 time=10 ms
Reply from 10.0.8.1: bytes=56 Sequence=2 ttl=254 time=1 ms
Reply from 10.0.8.1: bytes=56 Sequence=3 ttl=254 time=1 ms
Reply from 10.0.8.1: bytes=56 Sequence=4 ttl=254 time=10 ms
Reply from 10.0.8.1: bytes=56 Sequence=5 ttl=254 time=1 ms

--- 10.0.8.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/4/10 ms
[R2]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 10    Routes : 10

Destination/Mask    Proto Pre  Cost Flags        NextHop          Interface
10.0.4.0/24         Direct 0    0           D  10.0.4.254      GigabitEthernet0/0/1.1
10.0.4.254/32       Direct 0    0           D  127.0.0.1       GigabitEthernet0/0/1.1
10.0.4.255/32       Direct 0    0           D  127.0.0.1       GigabitEthernet0/0/1.1
10.0.8.0/24         Direct 0    0           D  10.0.8.254      GigabitEthernet0/0/1.3
10.0.8.254/32       Direct 0    0           D  127.0.0.1       GigabitEthernet0/0/1.3
10.0.8.255/32       Direct 0    0           D  127.0.0.1       GigabitEthernet0/0/1.3
127.0.0.0/8         Direct 0    0           D  127.0.0.1       InLoopBack0
127.0.0.1/32        Direct 0    0           D  127.0.0.1       InLoopBack0
127.255.255.255/32  Direct 0    0           D  127.0.0.1       InLoopBack0
255.255.255.255/32  Direct 0    0           D  127.0.0.1       InLoopBack0
```

Final Configuration

```
[R1]display current-configuration
[V200R007C00SPC600]
#
sysname R1
#
interface Ethernet0/0/0
ip address 10.0.4.1 255.255.255.0
#
```

```

ip route-static 0.0.0.0 0.0.0.0 10.0.4.254
#
user-interface con 0
authentication-mode password
set authentication password cipher %$$$dD#}P<HzJ;Xs%X>hOkm!.,+lq61QK`K6tl}cc-;k_o`C.+L,%$$$
user-interface vty 0 4
#
return

[R2]display current-configuration
[V200R007C00SPC600]
#
sysname R2
#
interface GigabitEthernet0/0/1
#
interface GigabitEthernet0/0/1.1
dot1q termination vid 4
ip address 10.0.4.254 255.255.255.0
arp broadcast enable
#
interface GigabitEthernet0/0/1.3
dot1q termination vid 8
ip address 10.0.8.254 255.255.255.0
arp broadcast enable
#
user-interface con 0
authentication-mode password
set authentication password cipher %$$$nRPL^hr2IXi7LHDID!/,.*%.8%h;3:,hXO2dk#ikaWl.*(,%$$$
user-interface vty 0 4
#
return
[R3]dis current-configuration
[V200R007C00SPC600]
#
sysname R3
#
interface Ethernet0/0/0
ip address 10.0.8.1 255.255.255.0
#
ip route-static 0.0.0.0 0.0.0.0 10.0.8.254
#
user-interface con 0
authentication-mode password
set authentication password cipher %$$$W|$)M5D}v@bY^gK\;>QR,.*d;8Mp>|+EU,:~D~8b59~..*g,%$$$
user-interface vty 0 4
#
return
[S1]display current-configuration
#
!Software Version V200R008C00SPC500
sysname S1
#
vlan batch 4 8
#
interface Ethernet0/0/1
port link-type access
port default vlan 4
#
interface GigabitEthernet0/0/2
port link-type trunk
port trunk allow-pass vlan 4 8
#
interface Ethernet0/0/2
port link-type access
port default vlan 8
#

```

```
user-interface con 0
user-interface vty 0 4
#
return
```

Lab 1-3 Configuring Layer 3 Switching

Learning Objectives

As a result of this lab section, you should achieve the following tasks:

- Configuration of VLAN interfaces.
- Establishment of VLAN routing on a single switch
- Perform VLAN routing over an Ethernet Trunk link.
- Perform dynamic routing between VLAN interfaces using OSPF.

Topology

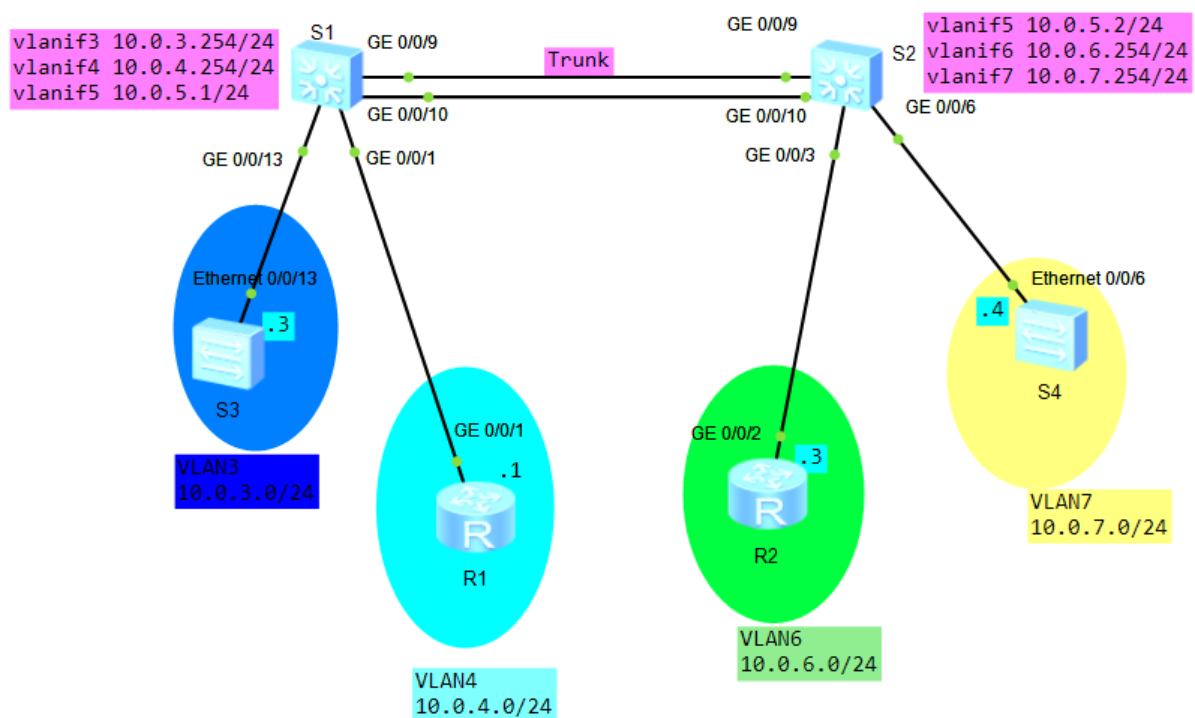


Figure 1.5 Layer 3 switching topology

Scenario

The introduction of layer three switches into the enterprise network opened up opportunities for streamlining the current VLAN routing configuration. The network administrator has been given the task to implement VLAN routing using only the layer three switches to support communication between the VLANs in the network as displayed in the topology. VLANs should be capable of inter VLAN communication. Additionally S1 and S2 are expected to communicate over a Layer 3 for which routing protocol support is required.

Tasks

Step 6 Preparing the environment

If you are starting this section with a non-configured device, begin here and then move to step 3. For those continuing from previous labs, begin at step 2.

Configure R1 with the address 10.0.4.1/24 on interface Gigabit Ethernet 0/0/1. Establish an Eth-Trunk between S1 and S2. Disable any unnecessary interfaces on S1 and S2 to S3 and S4.

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]interface GigabitEthernet 0/0/1
[R1-GigabitEthernet0/0/1]ip address 10.0.4.1 24
```

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
```

```
<Quidway>system-view
[Quidway]sysname S1
[S1]interface Eth-Trunk 1
[S1-Eth-Trunk1]mode lacp
[S1-Eth-Trunk1]port link-type trunk
[S1-Eth-Trunk1]port trunk allow-pass vlan all
[S1-Eth-Trunk1]quit
[S1]interface GigabitEthernet 0/0/9
[S1-GigabitEthernet0/0/9]eth-trunk 1
[S1-GigabitEthernet0/0/9]interface GigabitEthernet 0/0/10
[S1-GigabitEthernet0/0/10]eth-trunk 1
```

```
<Quidway>system-view
[Quidway]sysname S2
[S2]interface Eth-Trunk 1
[S2-Eth-Trunk1]mode lacp
[S2-Eth-Trunk1]port link-type trunk
[S2-Eth-Trunk1]port trunk allow-pass vlan all
[S2-Eth-Trunk1]quit
[S2]interface GigabitEthernet 0/0/9
[S2-GigabitEthernet0/0/9]eth-trunk 1
[S2-GigabitEthernet0/0/9]interface GigabitEthernet 0/0/10
[S2-GigabitEthernet0/0/10]eth-trunk 1
```

```
<Quidway>system-view
[Quidway]sysname S3
[S3]interface Ethernet 0/0/7
[S3-Ethernet0/0/7]shutdown
```

```
<Quidway>system-view
[Quidway]sysname S4
[S4]interface Ethernet 0/0/14
[S4-Ethernet0/0/14]shutdown
```

Step 7 Clean up the previous configuration

Remove the VLAN routing configuration and sub-interfaces on the devices.

```
[R1]undo ip route-static 0.0.0.0 0
```

```
[R2]undo interface GigabitEthernet 0/0/1.1
[R2]undo interface GigabitEthernet 0/0/1.3

[R3]interface GigabitEthernet 0/0/1
[R3-GigabitEthernet0/0/1]undo ip address
[R3-GigabitEthernet0/0/1]quit
[R3]undo ip route-static 0.0.0.0 0

[S1]undo vlan batch 4 8
Warning: The configurations of the VLAN will be deleted. Continue?[Y/N]:y
Info: This operation may take a few seconds. Please wait for a moment...done.
[S1]interface GigabitEthernet 0/0/2
[S1-GigabitEthernet0/0/2]undo port trunk allow-pass vlan 4 8
[S1-GigabitEthernet0/0/2]quit
[S1]interface GigabitEthernet 0/0/13
[S1-GigabitEthernet0/0/13]undo shutdown

[S2]interface GigabitEthernet0/0/6
[S2-GigabitEthernet0/0/6]undo shutdown
```

Re-enable the Eth-Trunk interface between S1 and S2

```
[S1]interface Eth-Trunk 1
[S1-Eth-Trunk1]undo shutdown

[S2]interface Eth-Trunk 1
[S2-Eth-Trunk1]undo shutdown
```

Step 8 Configure VLAN 3 through to VLAN 7 for S1 and S2

```
[S1]vlan batch 3 to 7
Info: This operation may take a few seconds. Please wait for a moment...done.

[S2]vlan batch 3 to 7
Info: This operation may take a few seconds. Please wait for a moment...done.
```

Verify that the VLANs have been created.

```
[S1]display vlan
The total number of vlans is : 6
...output omitted...
VID Type Ports
-----
1 common UT:GE0/0/1(U) GE0/0/2(D) GE0/0/3(U) GE0/0/4(U)
    GE0/0/5(U) GE0/0/6(D) GE0/0/7(D) GE0/0/8(D)
    GE0/0/11(D) GE0/0/12(D) GE0/0/13(D) GE0/0/14(D)
    GE0/0/15(D) GE0/0/16(D) GE0/0/17(D) GE0/0/18(D)
    GE0/0/19(D) GE0/0/20(D) GE0/0/21(U) GE0/0/22(U)
    GE0/0/23(U) GE0/0/24(D) Eth-Trunk1(U)
3 common TG:Eth-Trunk1(U)
4 common TG:Eth-Trunk1(U)
5 common TG:Eth-Trunk1(U)
6 common TG:Eth-Trunk1(U)
7 common TG:Eth-Trunk1(U)
...output omitted...
```

```
[S2]display vlan
The total number of vlans is : 6
...output omitted...
VID Type Ports
-----
1 common UT:GE0/0/1(U) GE0/0/2(D) GE0/0/3(U) GE0/0/4(U)
    GE0/0/5(U) GE0/0/6(D) GE0/0/7(D) GE0/0/8(D)
    GE0/0/11(U) GE0/0/12(U) GE0/0/13(U) GE0/0/14(D)
```

```

GE0/0/15(D)  GE0/0/16(D)  GE0/0/17(D)  GE0/0/18(D)
GE0/0/19(D)  GE0/0/20(D)  GE0/0/21(D)  GE0/0/22(D)
GE0/0/23(D)  GE0/0/24(D)  Eth-Trunk1(U)
3 common TG:Eth-Trunk1(U)
4 common TG:Eth-Trunk1(U)
5 common TG:Eth-Trunk1(U)
6 common TG:Eth-Trunk1(U)
7 common TG:Eth-Trunk1(U)
...output omitted...

```

Step 9 Set the Eth-Trunk link between S1 and S2 with PVID 5

Add interfaces Gigabit Ethernet 0/0/1 and 0/0/13 of S1 to VLAN 4 and VLAN 3 respectively. For S2, add interfaces Gigabit Ethernet 0/0/3 and G0/0/6 to VLAN 6 and VLAN 7 respectively.

```

[S1]interface Eth-Trunk 1
[S1-Eth-Trunk1]port trunk pvid vlan 5
[S1-Eth-Trunk1]quit
[S1]interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1]port link-type access
[S1-GigabitEthernet0/0/1]port default vlan 4
[S1-GigabitEthernet0/0/1]quit
[S1]interface GigabitEthernet 0/0/13
[S1-GigabitEthernet0/0/13]port link-type access
[S1-GigabitEthernet0/0/13]port default vlan 3

```

```

[S2]interface Eth-Trunk 1
[S2-Eth-Trunk1]port trunk pvid vlan 5
[S2-Eth-Trunk1]quit
[S2]interface GigabitEthernet 0/0/3
[S2-GigabitEthernet0/0/3]port link-type access
[S2-GigabitEthernet0/0/3]port default vlan 6
[S2-GigabitEthernet0/0/3]quit
[S2]interface GigabitEthernet 0/0/6
[S2-GigabitEthernet0/0/6]port link-type access
[S2-GigabitEthernet0/0/6]port default vlan 7

```

Run the **display vlan** command to view the configuration.

```

<S1>display vlan
The total number of vlans is : 6
...output omit...
VID Type  Ports
-----
1 common UT:GE0/0/2(D)  GE0/0/3(U)  GE0/0/4(U)  GE0/0/5(U)
      GE0/0/6(D)  GE0/0/7(D)  GE0/0/8(D)  GE0/0/11(D)
      GE0/0/12(D)  GE0/0/14(D)  GE0/0/15(D)  GE0/0/16(D)
      GE0/0/17(D)  GE0/0/18(D)  GE0/0/19(D)  GE0/0/20(D)
      GE0/0/21(U)  GE0/0/22(U)  GE0/0/23(U)  GE0/0/24(D)
      TG:Eth-Trunk1(U)
3 common  UT:GE0/0/13(U)
      TG:Eth-Trunk1(U)
4 common  UT:GE0/0/1(U)
      TG:Eth-Trunk1(U)
5 common  UT:Eth-Trunk1(U)
6 common  TG:Eth-Trunk1(U)
7 common  TG:Eth-Trunk1(U)
...output omit...

```

```

<S2>display vlan
The total number of vlans is : 6
...output omit...
VID Type  Ports
-----

```

```

1 common UT:GE0/0/1(U) GE0/0/2(D) GE0/0/4(U) GE0/0/5(U)
    GE0/0/6(D) GE0/0/7(D) GE0/0/8(D) GE0/0/11(U)
    GE0/0/12(U) GE0/0/13(U) GE0/0/14(D) GE0/0/15(D)
    GE0/0/16(D) GE0/0/17(D) GE0/0/18(D) GE0/0/19(D)
    GE0/0/20(D) GE0/0/21(D) GE0/0/22(D) GE0/0/23(D)
    TG:Eth-Trunk1(U)
3 common TG:Eth-Trunk1(U)
4 common TG:Eth-Trunk1(U)
5 common TG:Eth-Trunk1(U)
6 common UT:GE0/0/3(U)
    TG:Eth-Trunk1(U)
7 common UT:GE0/0/6(U)
    TG:Eth-Trunk1(U)
...output omit...

```

Step 10 **Configure gateway addresses for VLANs on S1 and S2**

Configure IP addresses for Vlanif3, Vlanif4, and Vlanif5 on S1, and for Vlanif5, Vlanif6, and Vlanif7 on S2.

```

[S1]interface Vlanif 3
[S1-Vlanif3]ip address 10.0.3.254 24
[S1-Vlanif3]interface Vlanif 4
[S1-Vlanif4]ip address 10.0.4.254 24
[S1-Vlanif4]interface Vlanif 5
[S1-Vlanif5]ip address 10.0.5.1 24

[S2]interface Vlanif 5
[S2-Vlanif5]ip address 10.0.5.2 24
[S2-Vlanif5]interface Vlanif 6
[S2-Vlanif6]ip address 10.0.6.254 24
[S2-Vlanif6]interface Vlanif 7
[S2-Vlanif7]ip address 10.0.7.254 24

```

Step 11 **IP addressing and default routes for R1, R2, S3 and S4**

IP addresses on a switch much be assigned to a Vlanif, where Vlanif1 is a common (untagged) Vlanif. Interfaces Ethernet 0/0/13 of S3 and Ethernet 0/0/6 of S4 should be associated with the common VLAN1. R1 should already be configured with the address 10.0.4.1/24.

```

[R1]ip route-static 0.0.0.0 0.0.0.0 10.0.4.254

[S3]interface Vlanif 1
[S3-Vlanif1]ip address 10.0.3.3 24
[S3-Vlanif1]quit
[S3]ip route-static 0.0.0.0 0.0.0.0 10.0.3.254

[R2]interface GigabitEthernet 0/0/2
[R2-GigabitEthernet0/0/2]ip address 10.0.6.3 24
[R2-GigabitEthernet0/0/2]quit
[R2]ip route-static 0.0.0.0 0.0.0.0 10.0.6.254

[S4]interface Vlanif 1
[S4-Vlanif1]ip address 10.0.7.4 24
[S4-Vlanif1]quit
[S4]ip route-static 0.0.0.0 0.0.0.0 10.0.7.254

```

Step 12 Test connectivity between VLAN 3 and VLAN 4

Test connectivity between S3 and R1.

```
<R1>ping 10.0.3.3
PING 10.0.3.3: 56 data bytes, press CTRL_C to break
Reply from 10.0.3.3: bytes=56 Sequence=1 ttl=254 time=37 ms
Reply from 10.0.3.3: bytes=56 Sequence=2 ttl=253 time=2 ms
Reply from 10.0.3.3: bytes=56 Sequence=3 ttl=253 time=10 ms
Reply from 10.0.3.3: bytes=56 Sequence=4 ttl=253 time=3 ms
Reply from 10.0.3.3: bytes=56 Sequence=5 ttl=253 time=2 ms

--- 10.0.3.3 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 2/10/37 ms
```

Test connectivity between R2 and R1.

```
<R1>ping 10.0.6.3
PING 10.0.6.3: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out

--- 10.0.6.3 ping statistics ---
5 packet(s) transmitted
0 packet(s) received
100.00% packet loss
```

The connectivity between R1 and R2 fails. Use the **tracert** command to troubleshoot the fault:

```
[R1]tracert 10.0.6.3
tracert to 10.0.6.3(10.0.6.3), max hops: 30 ,packet length: 40,press CTRL_C to break
 1 10.0.4.254 17 ms 4 ms 4 ms
 2 * * *
```

According to the command output, R1 has sent data packets to the destination address 10.0.6.3, but the gateway at 10.0.4.254 responds that the network is unreachable.

Check whether the network is unreachable on the gateway (S1).

```
[S1]display ip routing-table
Route Flags: R - relay, D - download to fib
```

```
-----
Routing Tables: Public
Destinations : 8      Routes : 8
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.3.0/24			Direct	0 0	D	10.0.3.254 Vlanif3
10.0.3.254/32			Direct	0 0	D	127.0.0.1 InLoopBack0
10.0.4.0/24			Direct	0 0	D	10.0.4.254 Vlanif4
10.0.4.254/32			Direct	0 0	D	127.0.0.1 InLoopBack0
10.0.5.0/24			Direct	0 0	D	10.0.5.1 Vlanif5
10.0.5.1/32			Direct	0 0	D	127.0.0.1 InLoopBack0
127.0.0.0/8			Direct	0 0	D	127.0.0.1 InLoopBack0

127.0.0.1/32 Direct 0 0 D 127.0.0.1 InLoopBack0

According to the command output, S1 does not have a route to the network segment 10.0.6.0 because the network segment is not directly connected to S1. In addition, no static route or dynamic routing protocol has been configured to advertise the routes.

Step 13 **Enable OSPF on S1 and S2**

```
[S1]ospf
[S1-ospf-1]area 0
[S1-ospf-1-area-0.0.0.0]network 10.0.0.0 0.255.255.255
```

```
[S2]ospf
[S2-ospf-1]area 0
[S2-ospf-1-area-0.0.0.0]network 10.0.0.0 0.255.255.255
```

After the configuration, wait until S1 and S2 exchange OSPF routes and complete the link state database, then view the resulting routing table of S1.

```
[S1]display ip routing-table
Route Flags: R - relay, D - download to fib
```

Routing Tables: Public

Destinations : 10 Routes : 10

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.3.0/24	Direct	0	0			Vlanif3
10.0.3.254/32	Direct	0	0	D	10.0.3.254	InLoopBack0
10.0.4.0/24	Direct	0	0			Vlanif4
10.0.4.254/32	Direct	0	0	D	10.0.4.254	InLoopBack0
10.0.5.0/24	Direct	0	0			Vlanif5
10.0.5.1/32	Direct	0	0	D	10.0.5.1	InLoopBack0
10.0.6.0/24	OSPF	10	2	D	10.0.5.2	Vlanif5
10.0.7.0/24	OSPF	10	2	D	10.0.5.2	Vlanif5
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0

S1 has learned two routes using OSPF. Test connectivity between R1 and R3.

```
[R1]ping 10.0.6.3
PING 10.0.6.3: 56 data bytes, press CTRL_C to break
Reply from 10.0.6.3: bytes=56 Sequence=1 ttl=253 time=11 ms
Reply from 10.0.6.3: bytes=56 Sequence=2 ttl=253 time=1 ms
Reply from 10.0.6.3: bytes=56 Sequence=3 ttl=253 time=10 ms
Reply from 10.0.6.3: bytes=56 Sequence=4 ttl=253 time=1 ms
Reply from 10.0.6.3: bytes=56 Sequence=5 ttl=253 time=1 ms
```

```
--- 10.0.6.3 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/4/11 ms
```

```
[R1]ping 10.0.7.4
PING 10.0.7.4: 56 data bytes, press CTRL_C to break
Reply from 10.0.7.4: bytes=56 Sequence=1 ttl=253 time=30 ms
Reply from 10.0.7.4: bytes=56 Sequence=2 ttl=252 time=2 ms
```

Reply from 10.0.7.4: bytes=56 Sequence=3 ttl=252 time=3 ms
Reply from 10.0.7.4: bytes=56 Sequence=4 ttl=252 time=2 ms
Reply from 10.0.7.4: bytes=56 Sequence=5 ttl=252 time=2 ms

--- 10.0.7.4 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 2/7/30 ms

Final Configuration

```
[R1]display current-configuration  
[V200R007C00SPC600]
```

```
#  
sysname R1  
#  
interface GigabitEthernet0/0/1  
ip address 10.0.4.1 255.255.255.0  
#  
ip route-static 0.0.0.0 0.0.0.0 10.0.4.254  
#  
return
```

```
[S1]display current-configuration  
!Software Version V200R008C00SPC500
```

```
#  
sysname S1  
#  
vlan batch 3 to 7  
#  
interface Vlanif3  
ip address 10.0.3.254 255.255.255.0  
#  
interface Vlanif4  
ip address 10.0.4.254 255.255.255.0  
#  
interface Vlanif5  
ip address 10.0.5.1 255.255.255.0  
#  
interface Eth-Trunk1  
port link-type trunk  
port trunk pvid vlan 5  
port trunk allow-pass vlan 2 to 4094  
mode lacp  
#  
interface GigabitEthernet0/0/1  
port link-type access  
port default vlan 4  
#  
interface GigabitEthernet0/0/9  
eth-trunk 1  
#  
interface GigabitEthernet0/0/10  
eth-trunk 1  
#  
interface GigabitEthernet0/0/13  
port link-type access  
port default vlan 3  
#  
ospf 1  
area 0.0.0.0  
network 10.0.0.0 0.255.255.255  
#  
return
```

```
[S2]display current-configuration
```

```
!Software Version V200R008C00SPC500
#
sysname S2
#
vlan batch 3 to 7
#
interface Vlanif5
 ip address 10.0.5.2 255.255.255.0
#
interface Vlanif6
 ip address 10.0.6.254 255.255.255.0
#
interface Vlanif7
 ip address 10.0.7.254 255.255.255.0
#
interface Eth-Trunk1
 port link-type trunk
 port trunk pvid vlan 5
 port trunk allow-pass vlan 2 to 4094
 mode lacp
#
interface GigabitEthernet0/0/3
 port link-type access
 port default vlan 6
#
interface GigabitEthernet0/0/6
 port link-type access
 port default vlan 7
#
interface GigabitEthernet0/0/9
 eth-trunk 1
#
interface GigabitEthernet0/0/10
 eth-trunk 1
#
ospf 1
 area 0.0.0.0
 network 10.0.0.0 0.255.255.255
#
return
```

```
[S3]display current-configuration
#
!Software Version V100R006C05
sysname S3
#
interface Vlanif1
 ip address 10.0.3.3 255.255.255.0
#
interface Ethernet0/0/7
 shutdown
#
 ip route-static 0.0.0.0 0.0.0.0 10.0.3.254
#
return
```

```
[S4]display current-configuration
#
!Software Version V100R006C05
sysname S4
#
interface Vlanif1
 ip address 10.0.7.4 255.255.255.0
#
interface Ethernet0/0/14
 shutdown
#
```

```
ip route-static 0.0.0.0 0.0.0.0 10.0.7.254  
#  
return
```